

Heart Attack Prediction using Deep Learning Approach based on ECG Signal



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ABSTRACT

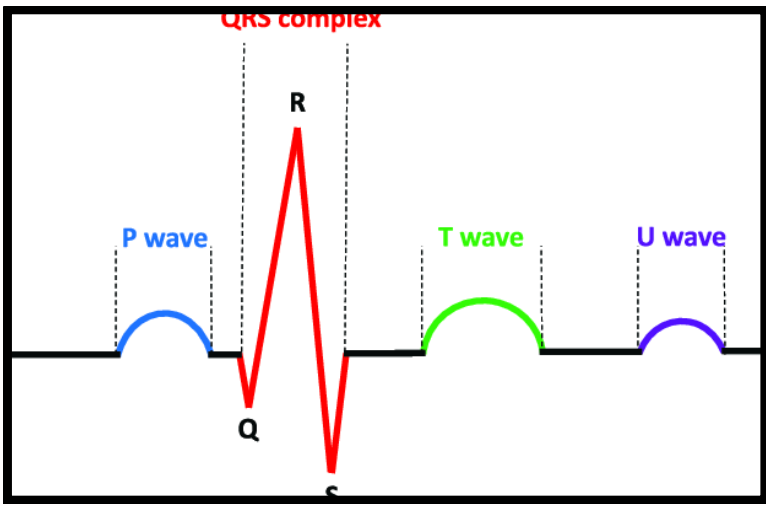
A CNN-based model designed to detect myocardial infarction from ECG images. Through image segmentation and advanced preprocessing, it achieves high accuracy and robustness. The system enables fast, precise, and non-invasive early detection to support clinical diagnosis.

INTRODUCTION

A CNN-based system that analyzes ECG images to detect myocardial infarction with high accuracy. It uses preprocessing for cleaner inputs and supports real-time, automated diagnosis, helping clinicians make faster, more reliable decisions while reducing manual interpretation time.

DATASET

ECG Waveform Explanation

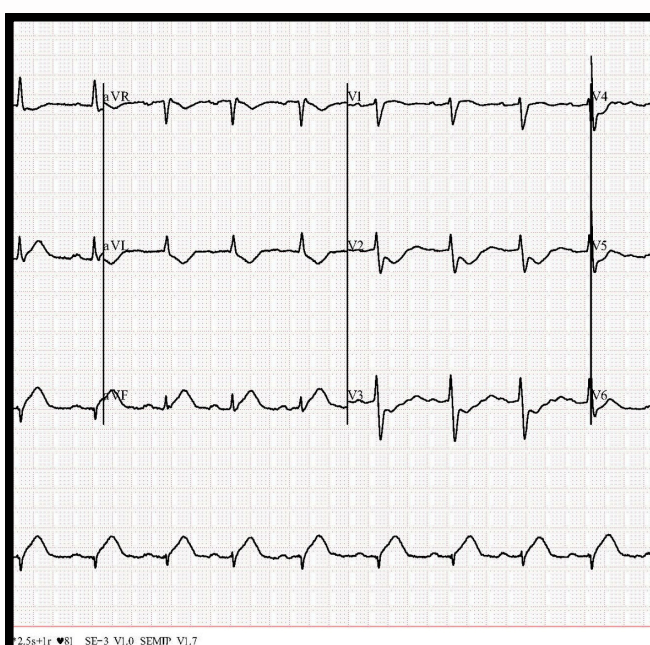
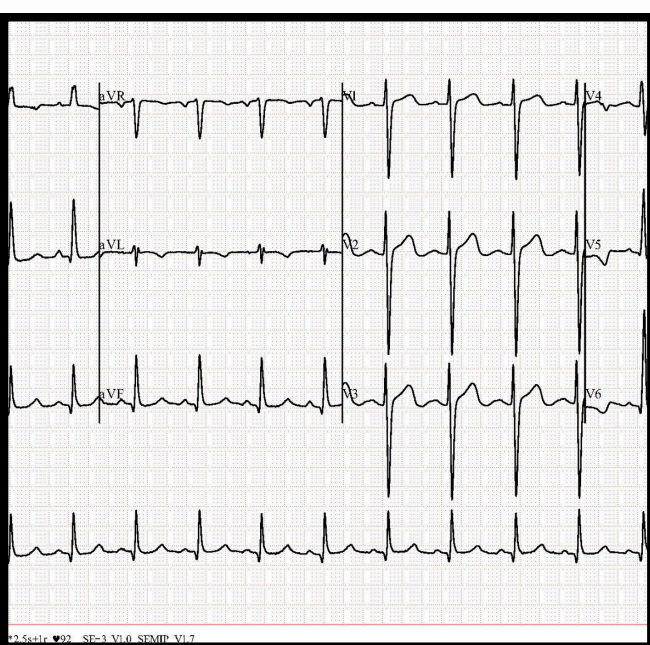


- **P wave:** Atrial depolarization (atria contraction)
- **QRS complex:** Ventricular depolarization (main diagnostic feature)
- **T wave:** Ventricular repolarization (recovery phase)
- **U wave:** Repolarization of Purkinje fibers (may not always appear)
- **Abnormalities in the QRS complex, ST segment, or T wave** often indicate a heart attack.

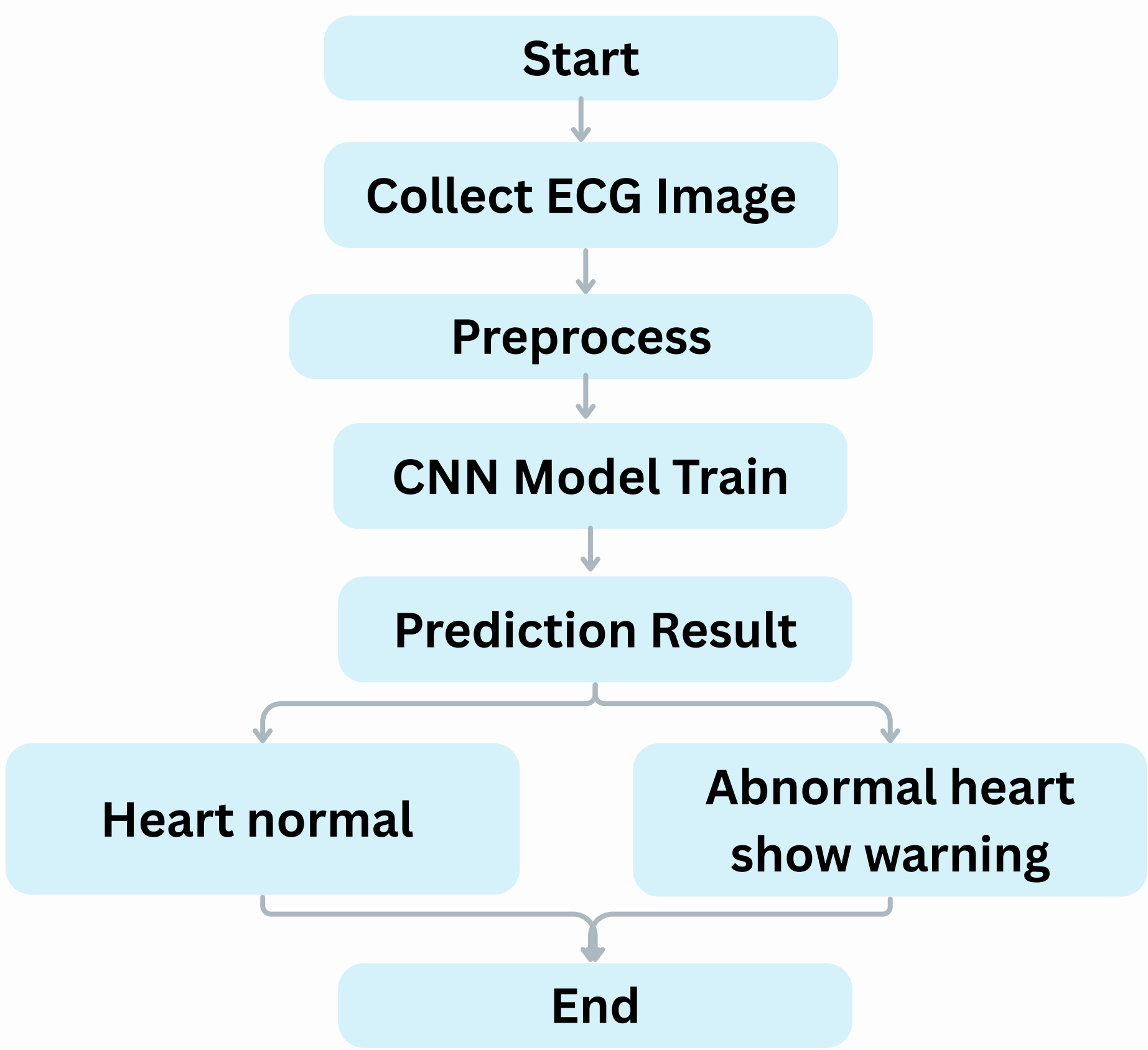
Normal

Myocardial Infarction

Abnormal Heart beat



METHODOLOGY



RESULT

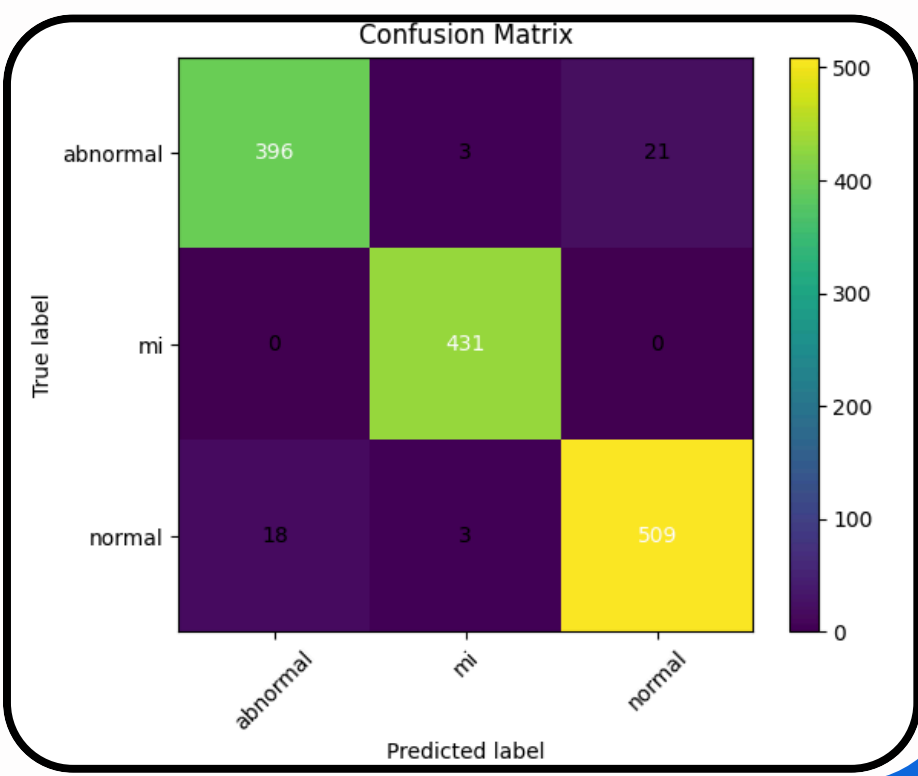
Classification Report

	precision	recall	f1-score	support
abnormal	0.96	0.94	0.95	420
mi	0.99	1.00	0.99	431
normal	0.96	0.96	0.96	530
accuracy			0.97	1381
macro avg	0.97	0.97	0.97	1381
weighted avg	0.97	0.97	0.97	1381

- Overall accuracy: 97%
- High precision, recall, and F1-score for all three classes
- Strong and consistent model performance

Confusion Matrix

- Most predictions correct across all classes
- Very few misclassifications observed
- Shows reliable class separation



CONCLUSION

In conclusion, this project effectively uses deep learning and image processing to predict heart attack risk from ECG data. With segmentation and preprocessing, it delivers an accurate, automated tool to aid early diagnosis and reduce mortality.

REFERENCE

- S. B. Patil and D. Kumaraswamy, "Extraction of significant patterns from heart disease warehouses for heart attack prediction," *IJCSNS International Journal of Computer Science and Network Security*, vol. 9, no. 2, pp. 228–235, 2009.
- S. Nashif, M. R. Raihan, M. R. Islam, and M. H. Imam, "Heart disease detection by using machine learning algorithms and a real-time cardiovascular health monitoring system." *World Journal of Engineering* ..